

Control Theory thru Simulation

> a feedback controller repeatedly performs:

Read current angle:

→ calculate error

→ choose control

→ step physics

→ repeat

1st controller: proportional control:

$$e = \theta_{\text{desired}} - \theta_{\text{current}}$$

$$\text{ctrl} = K_p e$$

↑
decides how strong
the controller reacts

→ small: gentle
not aggressive

→ large: likely overshoot
and oscillate

Using $K_p = 0.05$ control units per degree:

1. Error $+20^\circ \rightarrow$ what control command?

2. Error $-10^\circ \rightarrow$ what control command?

3. Error $+40^\circ$ gives a raw command of $+2$, but our actuator permits only $[-1, +1]$. What command should actually be sent?

1. $+1.0$

2. -0.5

3. will clip/saturate; so it returns upper bound $+1.0$

$$ctrl = \text{clip}(K_p e, ctrl_{min}, ctrl_{max})$$

Now an important limitation of P control.

At exactly the 30° target:

$$error = 0 \Rightarrow ctrl = 0$$

But gravity still pulls the pendulum downward. Therefore, a P controller may need a small persistent error to generate enough motor torque to oppose gravity. This is called steady-state error.

e.g. under gravity, P-controlled pendulum likely settle slightly below 30°

The pendulum would settle slightly below 30° .

At exactly 30° :

$$error = 30^\circ - 30^\circ = 0$$

Therefore, a P controller produces zero motor command. Gravity then pulls the pendulum in the negative direction, dropping it below the target.

For example, at 25° :

$$error = 30^\circ - 25^\circ = +5^\circ$$

That positive error produces positive motor torque. Eventually it can reach a point where:

$$\text{positive motor torque} = \text{negative gravity torque}$$

The pendulum then stops below the requested angle. The remaining difference is the steady-state error.

> P controller

feedback loop: the controller repeatedly reads the results of its previous action

Measure current angle
→ calculate error
→ calculate control command
→ advance physics
→ measure again

$$K_p = 2 \quad e = 30^\circ \rightarrow 0.524 \quad \rightarrow 2.0 \times 0.524 = 1.048 \xrightarrow{\text{clip}} +1.0$$